

Operating Conditions and Performance Standards

Robot Weld 01 (robot-weld-01) | Sensor 1: AXIS_4_SERVO_TORQUE

Machine	Robot Weld 01 (machine ID robot-weld-01, six-axis spot-welding robot, Body shop, site vit-lab)
Sensor	Sensor 1: AXIS_4_SERVO_TORQUE, "Axis 4 servo torque", sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE, type torque, unit Nm
Component / subsystem	Robot forearm (axis 4): axis 4 servo drive and gearbox; planned agent component tag robot_forearm_axis_4; suggested twin mesh 105-six-axis-welding-robot/arm-swing
Document title	Operating Conditions and Performance Standards
Document ID	RW01-S1-DOC-01
Revision / date	1.0 / 2026-10-08
Purpose	Defines what normal, degraded, warning, alarm and critical operation look like for Sensor 1 (axis 4 servo torque) so that a diagnostic agent can judge whether a reading, trend or pattern is acceptable, and how long an abnormal condition may persist before it is escalated.
Scope	Sensor 1 of Robot Weld 01 only (AXIS_4_SERVO_TORQUE, Nm). TOOL_CENTER_POINT_DEVIATION and WELD_GUN_TEMP are mentioned only as corroborating channels and are not documented here. Failure signatures are in Document 02, decision logic in Document 03, maintenance in Document 04, cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_SENSOR1.md (catalog limits, simulation equations, topics, tests).
[D]	Derived value: computed by the document authors from the repository's own simulation equations (noise-free unless stated).
[E]	Engineering reference: general industrial practice for robot servo-axis diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. An ESP32 gateway (or a software simulator) plays the robot controller and generates values from documented equations. The limits in this document that are tagged [P] are illustrative constants chosen for the demo in the repository, not OEM specifications. Values tagged [E] and [I] are not manufacturer-certified.

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1. Document Control and Retrieval Guide

1.1 Role of this document in the knowledge base

This is the reference for normal and abnormal values of Sensor 1. Retrieve it first when the question is whether a value, a window of values or a trend is acceptable. It contains the project limits as declared by the machine, the derived healthy signature, the engineering-reference extensions needed for diagnosis (watch, early-degradation and critical levels, feature thresholds, rate-of-change classes), the persistence and escalation rules, and the sensor limitations that can make a reading misleading.

Naming is fixed across the five documents: Sensor 1 = AXIS_4_SERVO_TORQUE = axis 4 servo torque, measured in Nm. The component is the robot forearm (axis 4): the axis 4 servo drive and gearbox. The only fault scenario in the repository for this machine is GEARBOX_WEAR, and Sensor 1 is its primary indicator.

Table 1. Question-to-section map for retrieval

Typical agent question	Answer location	Key numbers
Is this torque reading normal?	Sections 4, 5, 15	Healthy sample range about 3 to 23 Nm; cycle peak 21.75 Nm [D]
What value triggers a warning or an alarm?	Sections 6, 7, 16	warnHigh 26 Nm, alarmHigh 30 Nm [P]; proposed critical 34 Nm [I]
Is the machine drifting before any limit is crossed?	Sections 7, 8	Cycle mean above 13.5 Nm = watch; harmonic A6 above 0.4 Nm [I]
How long may an abnormal value persist?	Sections 9, 16	Peak-envelope rule: 2 cycles (16 s) to open, 2 cycles to clear [I]
Is this spike a transient?	Sections 8, 9	Healthy step between samples up to about 7 Nm in 0.5 s [D]
What can legitimately change the reading?	Sections 10, 11	Operating state, speed override, payload, temperature [E]
Can the sensor itself be wrong?	Section 12	Value is a controller estimate; accuracy not specified [P]
What does this imply diagnostically?	Section 17	GEARBOX_WEAR signature; see Documents 02 to 05

1.2 Provenance and confidence

Every number is tagged. Values tagged [P] come from firmware/catalog.json, the simulation equations, the MQTT contract and the server tests, and must not be changed without a repository change. Values tagged [D] are computed from those equations (the noise-free waveform, harmonic content and crossing points are exact; the percentages of samples in warn or alarm in a seeded run vary with the random seed). Values tagged [E] and [I] extend the project values so that the diagnostic agent has complete guidance. They are defensible engineering choices but they are not OEM values and should be recalibrated when real robot data exists.

2. Sensor and Component Overview

2.1 Identity of Sensor 1

Table 2. Sensor 1 identity

Field	Value	Basis
Machine	Robot Weld 01, machine ID robot-weld-01, type robot (typeCode 1), area Body shop, site vit-lab	[P]
Channel key (telemetry field)	AXIS_4_SERVO_TORQUE (upper-snake key inside telemetry field m)	[P]
Label / sensor type / unit	Axis 4 servo torque / torque / Nm	[P]
Sensor ID for twin bindings	ROBOT-WELD-01.AXIS_4_SERVO_TORQUE (MACHINE_ID.CHANNEL_KEY)	[P]
Channel position	Index 0 of the robot catalog (first channel in catalog order)	[P]
Declared range (clamp)	min 0 Nm, max 45 Nm	[P]
Resolution in telemetry	1 decimal (format %.1f)	[P]
Declared limits	warnHigh 26 Nm, alarmHigh 30 Nm; no low limits defined	[P]
Publish interval	500 ms (2 Hz), adjustable 250 to 5000 ms by command	[P]
Work cycle	8 s weld cycle; gun energised 4.4 to 6.0 s of each cycle (phase 0.55 to 0.75)	[P]
Scenarios	NORMAL, GEARBOX_WEAR (ramp 5 to 600 s, default 120 s)	[P]
Robot make, model, payload, reach, rated torque, gear ratio	Not specified in repository	[P]

2.2 Component and subsystem

Sensor 1 reports the torque delivered by the servo drive of robot axis 4, the forearm axis of the six-axis welding robot. The affected mechanical chain is: servo motor, axis 4 gearbox (the repository describes the fault scenario as a damaged gear mesh), the forearm structure and the wrist and tool flange that follow it. The twin suggestion for binding is the mesh 105-six-axis-welding-robot/arm-swing with display label Robot forearm (axis 4). The mesh is a suggestion: an operator chooses the actual mesh, and if the channel is unbound the agent must report needs_binding rather than invent a highlight.

Table 3. Component context

Item	Value	Note
Component	Robot forearm (axis 4)	Binding display label [P]
Agent component tag	robot_forearm_axis_4	Planned in Phase 5 blueprint; resolved through the sensor ID [P]
Upstream subsystem	Axis 4 servo drive and motor	Torque is computed here [E]
Downstream subsystem	Robot wrist and tool flange (tag tool_flange), spot weld gun (tag weld_gun)	Path accuracy is observed by Sensor 2 channel, not by Sensor 1 [P]
Mechanical load carried by axis 4	Forearm mass, wrist, flange, weld gun and cable dressing	Specific masses not specified in repository
Fault mechanism represented in the repository	Gearbox wear: damaged gear mesh	Only repository fault for this machine [P]

3. Measurement Principle and Units

3.1 What the value physically represents

The repository treats the value as a controller-internal state, not a bolt-on transducer. In a real plant a robot controller computes axis torque from motor phase current (torque is approximately the motor torque constant multiplied by the quadrature-axis current, multiplied by the gear ratio and gearbox efficiency) and publishes it through OPC UA or MTConnect to a gateway [E]. In this project the ESP32 gateway generates the number from an equation that stands in for that controller. The measurement principle is therefore documented in two layers: the simulated generation law [P] and the engineering reality it imitates [E].

3.2 Units and encoding

- Unit: Newton-metre (Nm). The value is the torque requested of axis 4 by the drive, positive in the model throughout (the catalog clamp floor is 0 Nm, so negative torque is not represented) [P].
- Encoding: one decimal place in the telemetry message field m.AXIS_4_SERVO_TORQUE, for example 17.5. Quantisation is therefore 0.1 Nm, small compared with the healthy noise standard deviation of 0.45 Nm [P][D].
- Sampling: one value every 500 ms. The cycle (8 s) is sampled 16 times. The highest frequency that can be observed is 1 Hz (Nyquist), so the fourth harmonic (0.5 Hz) and the sixth harmonic (0.75 Hz) are observable but anything faster, such as gear-mesh frequency, is not [D][E].
- Timestamps: epoch ms from the gateway clock; if synced is false or the stamp is more than two minutes from server time, the backend stamps its own receive time, which blurs the cycle phase [P].

Table 4. Generation law for NORMAL operation

Term	Expression	Meaning	Basis
Mean torque	13.0 Nm	Average load on axis 4 over the cycle	[P]
Fundamental	$7.0 \sin(2 \pi p)$, $p = (t \bmod 8 \text{ s})/8 \text{ s}$	Once-per-cycle swing from the forearm moving through its stroke (0.125 Hz)	[P]
Fourth harmonic	$2.2 \sin(8 \pi p)$	Four-times-per-cycle component (0.5 Hz)	[P]
Noise	0.45 x Gaussian	Standard deviation 0.45 Nm	[P]
Clamp	0 to 45 Nm	Values outside are limited	[P]

4. Healthy Operating Signature

4.1 Waveform of one weld cycle

Healthy torque is a repeating 8 s waveform. It has a noise-free minimum of 4.25 Nm, a noise-free maximum of 21.75 Nm at about 2.4 s into the cycle (phase about 0.30), a cycle mean of 13.0 Nm and noise of 0.45 Nm standard deviation [D]. The weld window (4.4 to 6.0 s) falls in the low-torque part of the cycle: noise-free torque there lies between 4.2 and 12.9 Nm with a mean of 7.4 Nm [D]. Therefore limit crossings caused by gearbox wear appear before the weld window, not during it.

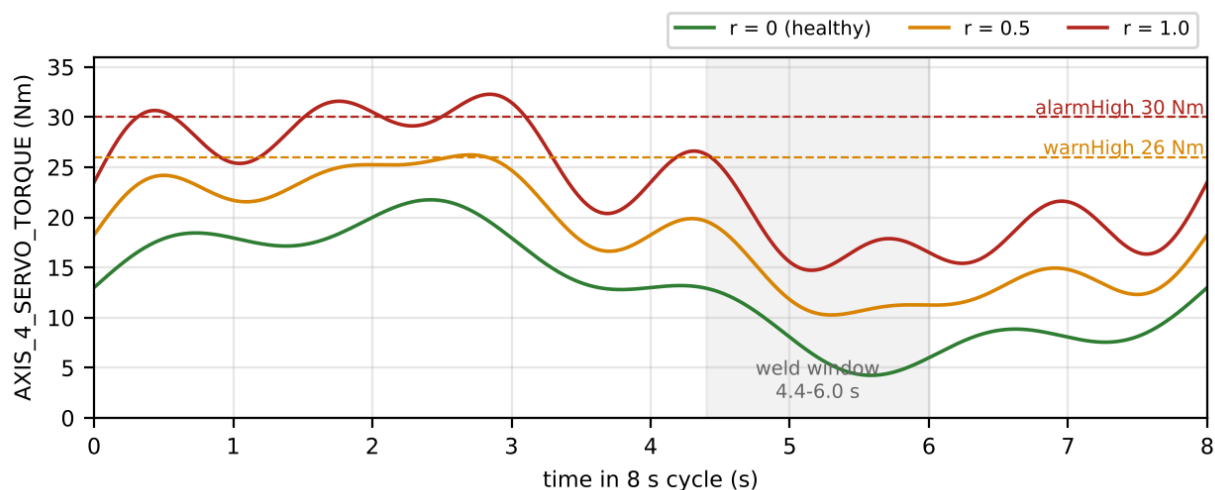


Figure 1. Noise-free torque over one 8 s cycle for healthy operation and two fault severities, with the project limits [D].

Table 5. Phase-resolved torque, noise-free (Nm) [D]

Time in cycle (s)	0	1	2	3	4	5	6	7
Healthy ($r = 0$)	13.0	17.9	20.0	17.9	13.0	8.1	6.0	8.1
Fault severity 0.5	18.2	21.7	25.2	24.7	18.2	11.8	11.2	14.8
Fault severity 1.0	23.5	25.4	30.5	31.4	23.5	15.6	16.5	21.6
Weld gun state	off	off	off	off	off (on from 4.4)	on	on to 6.0	off

4.2 Harmonic content

The healthy signal contains exactly two deterministic components: the cycle fundamental (7.0 Nm at 0.125 Hz) and the fourth harmonic (2.2 Nm at 0.5 Hz). Any energy at the sixth harmonic (0.75 Hz) is absent in a healthy axis and is the cleanest early indicator of gear-mesh damage in this dataset. See Figure 1 in Document 02 for the spectrum by severity.

Table 6. Healthy spectral lines of Sensor 1

Line	Frequency	Amplitude (Nm)	Healthy?	Basis
Mean (DC)	0	13.0	Yes	[P]
1x cycle	0.125 Hz	7.0	Yes	[P]
4x cycle	0.5 Hz	2.2	Yes	[P]
6x cycle	0.75 Hz	0 (estimator floor 0.1 to 0.2 with noise)	Absent when healthy	[D]
Broadband noise	all	sigma 0.45	Yes	[P]

4.3 Healthy sample statistics

Table 7. Healthy sample statistics over 600 s at 2 Hz (seeded runs) [D]

Statistic	Value (Nm)	Comment
Mean of all samples	13.0	Equals the model mean
Standard deviation of all samples	5.2	Dominated by the cycle swing, not by noise
Observed sample minimum	about 3.1 to 3.4	Noise-free minimum 4.25 minus noise tail
Observed sample maximum	about 22.6 to 23.1	Noise-free peak 21.75 plus noise tail
Per-cycle sample peak: mean / 99th percentile / maximum	21.7 / 22.7 / 23.0	Over 2000 simulated cycles
Margin from healthy per-cycle peak to warnHigh 26	about 3.3 Nm (mean) and 3.0 Nm (maximum)	Healthy operation never reaches warn (asserted by test over 600 s)
Standard deviation of the 8 s cycle mean (16 samples)	0.11	5-cycle mean: 0.05

5. Normal Operating Range and Expected Extremes

5.1 Normal operating range table

Table 8. Normal operating range for Sensor 1

Parameter	Lower limit	Normal range	Upper expected	Unit	Interpretation
Instantaneous torque, healthy	0 (clamp) [P]	4 to 22	23.1 (noise tail) [D]	Nm	Any sample inside 3 to 23.1 Nm is consistent with a healthy axis in production cycle
Noise-free cycle minimum	-	4.25	4.25 [D]	Nm	Reached at about 6.0 s of the cycle (end of weld window)
Noise-free cycle maximum	-	21.75	21.75 [D]	Nm	Reached at about 2.4 s; this is the anchor for the warning margin
Cycle mean (8 s, 16 samples)	12.5 [I]	12.7 to 13.3	13.5 [I]	Nm	Mean outside 12.5 to 13.5 indicates a baseline shift
Sixth-harmonic amplitude A6 (5 cycles)	0	0 to 0.2	0.2 [I]	Nm	Value above 0.4 means gear-mesh ripple is present
Noise standard deviation (cycle-to-cycle residual)	0.40 [I]	0.40 to 0.55	0.55 [I]	Nm	Increase indicates added roughness
Sample-to-sample step (0.5 s)	-	0 to 5.8 (99th percentile)	7.2 observed max [D]	Nm per 0.5 s	Cyclic movement produces steps up to about 7 Nm; steps above 8 Nm need context

5.2 Minimum and maximum expected values

Table 9. Minimum and maximum expected values by condition [D]

Condition	Noise-free minimum	Noise-free maximum	Mean	Highest plausible sample	Peak status
Healthy ($r = 0$)	4.25	21.75	13.0	23.1	normal
Severity 0.25	7.48	23.77	15.6	25.1	normal
Severity 0.48	10.07	26.01	18.0	27.6	warn begins
Severity 0.75	12.60	29.15	20.9	30.3	warn (alarm on rare samples)
Severity 0.82	13.20	30.02	21.6	32 (noise)	alarm begins
Severity 1.00 (full)	14.72	32.28	23.5	34.0	alarm

Note the asymmetry: wear raises the minimum as well as the maximum. At full severity the cycle minimum rises from 4.25 to 14.72 Nm, so a high cycle minimum is a robust indicator of a baseline shift that is not affected by the sparse-peak behaviour of the warning limit.

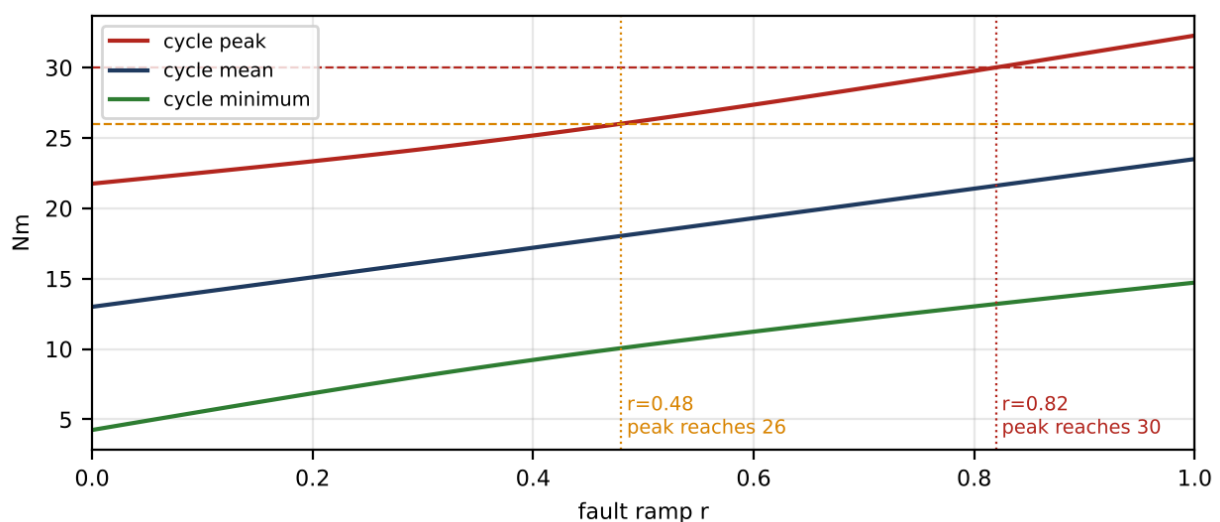


Figure 2. Cycle mean, minimum and peak of Sensor 1 against fault severity r ; the project limits are crossed by the peak only [D].

6. Threshold System

6.1 Project limits and what they mean

The machine declares two upper limits in its birth message: warnHigh 26 Nm and alarmHigh 30 Nm [P]. A sample is classified warn when its value reaches 26 Nm and alarm when it reaches 30 Nm (comparison is greater-than-or-equal; alarm wins over warn). No low limits are declared for this channel, so a low torque is never flagged by the platform. The classification is applied to each sample independently, with no hysteresis, delay or latching [P].

Consequence for interpretation: because the signal is cyclic and the limits are per sample, status flickers once per cycle and the marker is not steady. The warning limit is crossed only near the cycle peak (about 2 to 3.5 s into the cycle). Persistence must therefore be judged on the per-cycle peak envelope, not on consecutive raw samples.

6.2 Five-level threshold model used by this knowledge base

The task of a diagnostic agent needs more levels than the two the machine declares. The levels below keep the project warn and alarm limits unchanged and add a lower limit, a watch level and a critical level around them. Each level states what it means and what must be done.

Table 10. Threshold definitions for Sensor 1 torque (per sample)

Level	Value (Nm)	Basis	Meaning	Required response
Lower limit / minimum	0 (clamp floor); expected healthy minimum 3.1	[P][D]	Below 3.0 Nm in production cycle is outside the healthy envelope	Treat as possible sensor or signal fault; see Section 12 and Document 02 (sensor faults)
Normal	3.1 to 23.1	[D]	Healthy production cycle	None
Watch (advisory)	23.2 to 25.9, or feature watch levels in Section 7	[D][I]	Above anything a healthy axis produces but below the warn limit	Flag, increase trend monitoring, compare with baseline
Warning	26.0 to 29.9	[P]	warnHigh reached; status warn, amber on twin	Open investigation, schedule inspection at next planned stop
Alarm	30.0 to 33.9	[P]	alarmHigh reached; status alarm, red pulsing on twin	Intervention required; reduce duty, plan stop, do not run to failure
Critical	34.0 and above, or alarm persisting beyond escalation time	[I]	Above the full-severity peak of the modeled fault (32.3 Nm) plus noise margin; outside the modeled envelope	Immediate escalation; stop at a safe point; specialist inspection
Catalog maximum	45	[P]	Values clamp here	Any sustained value near 45 Nm implies saturation or sensor fault

6.3 Why 34 Nm is proposed as the critical threshold

The modeled fault saturates at a noise-free peak of 32.28 Nm at severity 1 [D]. Adding three noise standard deviations at full severity ($3 \times 1.125 = 3.4$ Nm) to the plateau of the worst case gives about 35.7 Nm as an extreme sample; a threshold of 34 Nm is crossed by roughly the upper tail of full-severity cycles (the 95th percentile of the per-cycle peak is 33.4 Nm [D]). A sample at or above 34 Nm therefore indicates a cycle at or beyond the maximum modeled severity and is the proposed trigger for immediate escalation [I]. If the real robot has a rated or peak torque for axis 4, that figure replaces this proposal.

7. Feature-Based Thresholds

Sample limits alone detect wear late: the warning limit is only reached at severity 0.48, when the cycle mean has already risen by 5 Nm. The agent should therefore compute four window features and compare them with the thresholds below. Features are evaluated over a window of at least one full cycle (16 samples) and preferably five cycles (40 s, 80 samples) aligned to the 8 s cycle [I]. Severity index $S = (M - 13.0) / 10.5$ converts a cycle mean M into the equivalent fault severity; it is computed from telemetry and must not be confused with the simulator ground truth, which is hidden from the agent [D].

Table 11. Feature thresholds for Sensor 1 (5-cycle window)

Feature	Healthy	Watch	Early degradation	Warning-equivalent	Alarm-equivalent	Critical-equivalent	Unit
Cycle mean M	12.5 to 13.5	13.5 to 15.0	15.0 to 18.0	18.0 to 21.6	21.6 to 23.5	23.5 and above	Nm
Severity index S	-0.05 to 0.05	0.05 to 0.19	0.19 to 0.48	0.48 to 0.82	0.82 to 1.0	1.0 and above	-
Cycle minimum (mean of per-cycle sample minimum)	about 4.3	5.0 to 6.5	6.5 to 10.0	10.0 to 13.2	13.2 to 15.0	15.0 and above	Nm
Mean of per-cycle sample peak	21.7	22.5 to 23.5	23.5 to 26.0	26.0 to 29.7	29.7 to 32.0	32.0 and above	Nm
Sixth-harmonic amplitude A6	0 to 0.2	0.4 to 0.6	0.6 to 1.5	1.5 to 2.5	2.5 to 3.0	3.0 and above	Nm
Noise standard deviation (after removing mean and cycle template)	0.40 to 0.55	0.55 to 0.62	0.62 to 0.77	0.77 to 1.0	1.0 to 1.125	1.125 and above	Nm
Fraction of cycles with a sample at or above 26 Nm	0 percent	0 percent	0 to 9 percent	41 to 100 percent	100 percent	100 percent	-
Fraction of cycles with a sample at or above 30 Nm	0	0	0	0 to 5 percent	31 to 100 percent	100 percent	-

The feature bands are derived from the model equations at severity values 0.05, 0.19, 0.48, 0.82 and 1.0 [D]; the boundaries between bands are proposals [I]. The bands overlap in meaning, not in value: M is the most robust single feature because noise averages out (standard deviation 0.05 Nm over five cycles), A6 is the most specific because it is absent in a healthy axis, and the peak-based features connect directly to the project limits.

7.1 Companion channel on the same machine

Table 12. TOOL_CENTER_POINT_DEVIATION as corroboration (not Sensor 1)

Quantity	Healthy	Warn limit	Alarm limit	Gearbox wear behaviour
TCP deviation mean (mm)	about 0.104 [D]	0.25 [P]	0.5 [P]	Rises as $0.104 + 0.55 r$; crosses 0.25 at r about 0.27 and 0.5 at r about 0.72 [D]
Order of crossings	-	TCP before torque	TCP before torque	TCP warns at r 0.27, torque peak warns at r 0.48; TCP alarms at r 0.72, torque peak alarms at r 0.82 [D]

A rising torque mean accompanied by a rising TCP deviation supports GEARBOX_WEAR; a rising torque mean with unchanged TCP deviation points away from the modeled fault (see Document 02).

8. Rate-of-Change Limits

Two different rates matter. The within-cycle rate is the normal motion of the robot and must not be mistaken for a fault. The trend rate is the speed at which a feature (mean, A6, noise) moves from one window to the next and carries the diagnostic meaning.

Table 13. Within-cycle rate of change (noise included, 2 Hz samples) [D]

Condition	Maximum noise-free slope (Nm/s)	99th percentile step per 0.5 s (Nm)	Maximum observed step per 0.5 s (Nm)	Standard deviation of step (Nm)
Healthy	12.4	5.8	7.2	3.0
Severity 0.5	19.5	not computed	not computed	-
Severity 1.0	26.5	11.7	14.5	5.1

IF a single sample-to-sample step exceeds 8 Nm AND the next sample returns to within 6 Nm of the cycle template for that phase THEN classify it as a transient or glitch; IF steps larger than 8 Nm occur repeatedly at the same cycle phase over 3 consecutive cycles THEN treat as a recurring feature of the process and evaluate it against Section 9.

Table 14. Trend rate of change of cycle mean M: interpretation classes

Class	Rate of M	Time to move 1 Nm	Typical meaning	Basis
Step change	1 Nm or more within one cycle (8 s)	at most 8 s	Process change, payload change, program change, collision, brake or enable event, sensor offset; not gradual wear	[E]
Fast ramp	0.05 to 1 Nm per minute	1 to 20 min	Simulator fault injection with rampSec 5 to 120 s (0.0875 Nm/s at 120 s equals 5.25 Nm per min); real-world analogue is acute damage or lubrication loss	[D][E]
Slow ramp	0.5 to 5 Nm per hour	12 min to 2 h	Fast-developing degradation; simulator rampSec 600 s is 1.05 Nm per min	[D][E]
Gradual drift	0.02 to 0.5 Nm per day	2 to 50 days	Typical progressive gearbox wear in operation	[E][I]
Seasonal or thermal	less than 0.5 Nm per hour, reversible	-	Warm-up or ambient dependence, returns when conditions return	[E]

Simulator ramp arithmetic [D]: the mean rises by 10.5 Nm over rampSec, so $dM/dt = 10.5 / \text{rampSec}$. For rampSec = 120 s this is 0.0875 Nm/s; for 600 s it is 0.0175 Nm/s; for 5 s it is 2.1 Nm/s. With the default ramp the first warn peak appears after about 58 s and the first alarm peak after about 98 s [D].

9. Transient and Sustained Behaviour

9.1 Normal transient behaviour

Table 15. Expected transient behaviour of Sensor 1

Transient	Typical appearance	Duration	Normal?	How to recognise
Cycle peak	Rise to about 21.75 Nm at about 2.4 s	about 1 s above 20 Nm	Yes	Repeats every 8 s at the same phase
Weld window trough	Fall to about 4 to 8 Nm near 5 to 6.5 s	about 2 s	Yes	Coincides with gun energised 4.4 to 6.0 s
Noise spike	Single sample 2 to 3 sigma (0.9 to 1.4 Nm) away from template	1 sample (0.5 s)	Yes	Next sample returns; template residual only
Isolated large outlier	Residual above 2.5 Nm (about 5.5 sigma) in one sample	1 sample	Not expected in the model	Treat as glitch if isolated; see Section 12
Start of fault injection	Mean starts to rise from the baseline, ripple appears	seconds to minutes	No	Not a transient; it is the onset of GEARBOX_WEAR
Return to NORMAL after reset	Mean returns to 13.0 Nm at once	one cycle	Yes after command NORMAL or reboot	Ack message shows scenario NORMAL
Telemetry gap	Missing samples, seq jumps	seconds	Link issue, not a torque effect	seq gaps counted as packet loss; new bootId with seq 0 is a reboot

9.2 Sustained abnormal behaviour

Table 16. Sustained abnormal patterns and thresholds for persistence [I]

Pattern	Detection test	Minimum persistence to classify	Classification
Mean shift	M above 13.5 Nm	5 cycles (40 s) simulator; 1 week real [I]	Baseline shift; evaluate fault candidates
Warning peaks	Per-cycle peak at or above 26 Nm	2 consecutive cycles (16 s)	Warning condition opened
Alarm peaks	Per-cycle peak at or above 30 Nm	2 consecutive cycles (16 s)	Alarm condition opened
Critical sample	Any sample at or above 34 Nm	1 sample, confirmed by the next cycle	Critical escalation
Ripple present	A6 at or above 0.4 Nm over 5 cycles	5 cycles (40 s)	Gear-mesh ripple confirmed
Clearing	Peak envelope below limit minus 1 Nm	2 consecutive cycles (16 s)	Condition cleared (hysteresis)

The planned Phase 5 detector in the repository uses a 10 s dwell for threshold opening and clearing, a rolling z-score above 3 for 30 s for drift, a sustained positive slope of the 60 s mean over 10 minutes, and a 10 minute cooldown with one open investigation per machine [P, planned, not built]. Since the 8 s cycle means that raw samples above the limit appear for at most about 1.4 s per cycle even at full severity [D], a dwell of 10 s on raw samples would never be satisfied. The proposal above applies the same dwell to the per-cycle peak envelope: two cycles is 16 s, which is consistent with the repository's intention of a 10 s dwell.

10. Operating States and Load-Dependent Behaviour

The repository simulator has no idle, startup or shutdown state: whenever the gateway is online, the 8 s weld cycle runs continuously and the Gaussian noise is always present. The states below are engineering-reference extensions [E] that the agent needs when interpreting a real robot, and each is marked as not modeled by the simulator. A diagnostic conclusion must never be drawn from torque in a state other than production cycle without applying the correction in this table.

Table 17. Operating-state comparison

State	Description	Sensor 1 behaviour	Typical level (Nm)	Use for diagnosis?	Modeled in repository?
Production cycle (steady)	8 s weld cycle at programmed speed and payload	8 s waveform with mean 13.0, swing 7.0, harmonic 2.2	3 to 23 healthy	Yes: the reference state for every threshold	Yes [P]
Warm-up (first minutes or first cycles after long stop)	Grease cold, higher viscous friction	Mean shifted up, relaxing over minutes [E]	+0.5 to +1.5 Nm above steady mean [I]	Only after warm-up	No
Idle, servo enabled, stationary	Axis holds posture against gravity	Nearly constant value equal to holding torque of the posture; low noise	Posture dependent, illustrative 2 to 8 [I]	No (cycle features undefined)	No
Brake engaged / servo off	Holding by mechanical brake	Torque about 0 or not reported	0	No	No
Jog or teach	Manual slow motion	Irregular, not cyclic, low speed	0 to 15 [I]	No	No
Speed override below 100 percent	Production at reduced speed	Same shape stretched in time, lower dynamic component, same gravity component [E]	Peak lower than 21.75	Only with scaling by speed	No
Payload or tool change	Different forearm payload	Mean and swing change, ripple does not appear [E]	Mean shift of several Nm	Re-baseline	No
Fault state GEARBOX_WEAR	Ramp r from 0 to 1	Mean +10.5 r, ripple 3.0 r at 0.75 Hz, noise x (1 + 1.5 r)	mean 13.0 to 23.5	Yes	Yes [P]
Gateway offline / stale	No telemetry for max(5 s, 4 x interval)	No data	-	No	Yes [P]

10.1 Startup and shutdown behaviour

- Gateway startup (modeled): the board connects to Wi-Fi, synchronises its clock by SNTP, publishes birth, status online and then telemetry. The first telemetry messages may carry synced false and be stamped by the server on receipt. Do not compute cycle-aligned features until at least one full cycle (8 s) of synced samples exists [P].
- Robot startup (not modeled): after power-up and brake release the axis holds posture; treat the first production cycle after enable as warm-up and exclude it from baseline [E].
- Shutdown (modeled at gateway level): a power cut produces status offline about 15 s later through the Last Will; the last samples before that must not be read as a drop in torque [P].
- Stop at end of cycle (not modeled): torque falls to the holding value and the cyclic features disappear. A flat trace close to the holding torque is idle, not a failed sensor, if the status is online and the robot program is not running [E].

10.2 Load dependence

Axis 4 torque scales with the mass carried beyond the axis, the acceleration of the motion and the orientation of the axis relative to gravity [E]. In the simulator the load is fixed, and therefore the mean of 13.0 Nm and the cycle swing of 7.0 Nm are constants of the healthy baseline. On a real machine a payload change shifts the mean and the swing in proportion, whereas gearbox wear shifts the mean and adds the ripple but does not change the swing amplitude (the 7.0 Nm fundamental is unchanged in the model [D]). The agent should therefore use the pair mean-shift plus ripple, not mean-shift alone, to attribute a change to wear.

Table 18. Load-dependent interpretation

Change in operating load	Effect on mean M	Effect on cycle swing	Effect on A6	Effect on noise	Reading
Heavier tool or payload [E]	Up in proportion	Up in proportion	None	Little	Process change, not wear
Higher speed or acceleration [E]	Little	Up (dynamic part)	None	Little	Program change
Lower speed override [E]	Little	Down	None	Little	Operator action
GEARBOX_WEAR [P]	Up by 10.5 r	Unchanged (7.0)	Up by 3.0 r	Up by factor 1 + 1.5 r	Fault

11. Environmental Influences

The repository models no environmental effect on torque: WELD_GUN_TEMP has an ambient of 32 degC but it does not feed back into the torque equation and GEARBOX_WEAR does not change gun temperature [P]. The influences below describe what a real robot would show and how to avoid misreading it [E].

Table 19. Environmental and operational influences on Sensor 1

Influence	Mechanism	Direction and size [E][I]	Reversible?	Diagnostic handling
Ambient temperature (cold start)	Grease viscosity raises friction	+0.5 to +1.5 Nm for a cold gearbox	Yes, within minutes of running	Exclude warm-up cycles
Sustained high ambient or hot weld cell	Thinner lubricant, thermal expansion of gear backlash	Small decrease in friction, possible increase of ripple at high wear	Yes	Compare at similar temperature
Gun temperature (weld heat radiation)	Heat near flange and wrist	Negligible on axis 4 torque [E]	Yes	Use gun temperature only to exclude a thermal event, as in the simulator
Supply voltage dip	Drive limits current, torque estimate clipped or noisy	Dips in torque reading during acceleration	Yes	Cross-check drive fault log
Contamination (spatter, dust)	Cable dressing or seal drag	Slow rise of mean and noise	No (needs cleaning)	Inspect dress pack and seals
Humidity and washdown	Seal degradation, corrosion	Long-term	No	Inspect seals
Vibration from adjacent equipment (press)	Mechanical coupling through floor	Added noise at low frequency	Yes	Check whether the press is running
Payload / tooling change	Mass change	Mean and swing change	Yes	Re-baseline after the change
Program or path change	Different accelerations	Waveform shape changes	Yes	Use program ID in context

12. Sensor Limitations and Measurement Uncertainty

12.1 Limitations

- Not an independent transducer: the number is computed inside the controller (modeled by an equation). A failure of the drive, of the current measurement or of the controller model biases the reading without any mechanical cause [E].
- Torque is an estimate, not a gear-tooth measurement: gear-mesh frequencies and bearing defect frequencies are far above the 1 Hz observable limit at 2 Hz sampling, so only low-order cycle harmonics are visible [D][E].
- Per-sample limits and a cyclic signal: a sample crosses a limit only at the cycle peak, so severity must be inferred from features rather than from the proportion of time in warn [D].
- Quantisation of 0.1 Nm, noise 0.45 Nm healthy, clamp at 0 and 45 Nm: saturation hides values above 45 Nm [P].
- Cycle phase is not published: the agent must infer the phase from the 8 s periodicity or from the weld gun trace; using server receive time instead of gateway time degrades alignment [P].
- Accuracy, calibration, rated torque and gear ratio are not specified in the repository [P].

12.2 Measurement uncertainty

Table 20. Uncertainty budget for Sensor 1 features [D][E]

Quantity	Source	1 sigma	Effect on diagnosis
Single sample	Noise	0.45 Nm healthy, 1.125 Nm at full severity	Single sample can differ from the template by about 1 Nm; do not diagnose from one sample
Quantisation	1 decimal	0.03 Nm (0.1 / sqrt(12))	Negligible
Cycle mean over 1 cycle	16 samples	0.11 Nm (healthy), 0.28 Nm (full severity)	Detects 1 Nm shifts with about 9 sigma confidence
Cycle mean over 5 cycles	80 samples	0.05 Nm (healthy)	Detects shifts of 0.3 Nm (S 0.03)
A6 over 1 cycle	16 samples	estimator floor 0.20 mean, 95th percentile 0.39 Nm (healthy)	Reliable only above 0.4 Nm
A6 over 5 cycles	80 samples	floor 0.09, 95th percentile 0.18 Nm (healthy)	Reliable above 0.2 Nm
Gateway clock and phase	SNTP, receive time fallback	unspecified	Phase error blurs A6; use gateway ts when synced true
Real-sensor accuracy	Controller estimate	Not specified in repository (engineering assumption of 2 to 5 percent of rated torque [E])	Add margin to thresholds on a real robot

13. Baseline Establishment and Data Quality

13.1 Building the healthy baseline

Every feature threshold in this document is anchored to the healthy baseline of the specific robot. In the simulator the baseline is known exactly; on a real robot it must be measured. The procedure below is the engineering-reference method [E][I] and is also what the diagnostic agent should apply when a baseline record is requested.

Table 21. Baseline procedure

Step	Action	Acceptance criterion	Output
1	Confirm gateway status online, synced true, no seq gaps in the window	No gap longer than 2 intervals (1 s); synced true on every message	Clean window
2	Confirm production cycle at 100 percent speed override, payload unchanged, warm-up complete	At least 3 cycles since warm-up (about 24 s in the simulator, 15 to 30 min on a real robot [E])	Valid operating state
3	Collect at least 20 consecutive cycles (160 s, 320 samples)	Cycle length 8.0 s within 1 percent	Baseline sample set
4	Fold samples by cycle phase into 16 bins of 0.5 s and compute the mean per bin	Bin mean standard error below 0.15 Nm (sigma 0.45 / sqrt 20)	Cycle template
5	Compute baseline M, per-cycle peak, per-cycle minimum, noise sigma (residual to template), A6 (projection onto the 6th harmonic)	M within 12.5 to 13.5 Nm and A6 below 0.2 Nm for the simulated robot	Baseline feature vector
6	Store baseline with timestamp, program ID and payload	-	Baseline record

For the simulated robot the expected baseline is M = 13.0 Nm, template peak 21.75 Nm at 2.4 s, minimum 4.25 Nm, sigma 0.45 Nm and A6 near 0 [D]. If a measured baseline differs materially from these values, either the robot is not in the NORMAL scenario (a fault may already be injected) or the sample window is not in the healthy state.

13.2 Data-quality preconditions

Table 22. Data-quality conditions that invalidate threshold evaluation

Condition	How it appears in the platform	Effect on Sensor 1 interpretation	Required action
Device stale	No telemetry for max(5 s, 4 x intervalMs)	No current value; last value is old	Do not evaluate; report stale
Device offline	status offline, Last Will after about 15 s	No data	Report offline; check power and network
Sequence gaps	seq jumps in telemetry	Missing samples distort peak estimates	Mark window incomplete if more than 10 percent of samples are missing
Reboot	New bootId, seq restarts at 0	Phase continuity lost	Restart cycle alignment; do not count as packet loss
Clock not synced	synced false or ts more than 2 minutes from server	Server stamps receive time; jitter blurs phase	Use mean, peak and noise features only; skip A6
Interval changed	cmd/interval changes the period (250 to 5000 ms)	Samples per cycle changes (32 at 250 ms, 1.6 at 5000 ms)	Recompute feature windows; at 5000 ms ripple at 0.75 Hz cannot be resolved
Identity conflict	Board and simulator both announcing robot-weld-01	Two sources interleave	Stop; resolve identity before any diagnosis
Schema-invalid message	Dropped and counted by the backend	Silent loss of samples	Check ingestion counters on /health

13.3 Time to threshold under the simulator ramp

When a fault is injected the ramp time rampSec (5 to 600 s, default 120 s) sets how quickly the thresholds are reached. The values below follow from $r = t / \text{rampSec}$ and the crossing points $r = 0.48$ (first warn peak) and $r = 0.82$ (first alarm peak) [D]. They let the agent judge whether an observed progression is consistent with the simulator and what lead time the feature thresholds give over the project limits.

Table 23. Time to reach thresholds by ramp length [D]

rampSec	Watch (M 13.5 Nm, r 0.05)	M 15 Nm (r 0.19)	First warn peak (r 0.48)	First alarm peak (r 0.82)	Full severity (r 1)	Lead of M 15 Nm over warn
5 s	0.2 s	0.95 s	2.4 s	4.1 s	5 s	1.4 s
30 s	1.5 s	5.7 s	14.4 s	24.6 s	30 s	8.7 s
60 s	3.0 s	11.4 s	28.8 s	49.2 s	60 s	17.4 s
120 s (default)	6 s	22.8 s	57.6 s	98.4 s	120 s	34.8 s
300 s	15 s	57 s	144 s	246 s	300 s	87 s
600 s	30 s	114 s	288 s	492 s	600 s	174 s

13.4 Staged definition of gearbox degradation as seen by Sensor 1

The stage table gives the agent a single place to convert any feature vector into a named stage. Stage boundaries correspond to the feature thresholds in Section 7 and to the project limits.

Table 24. Degradation stages of the axis 4 gear train as observed in Sensor 1

Stage	Name	Severity S	Cycle mean M (Nm)	Cycle peak (Nm)	A6 (Nm)	Noise sigma (Nm)	Platform status	Recommended stance
0	Healthy	0 to 0.05	12.5 to 13.5	21.75	0 to 0.2	0.45	normal	Routine monitoring
1	Incipient	0.05 to 0.19	13.5 to 15.0	22.3 to 23.7	0.4 to 0.6	0.48 to 0.58	normal	Watch; trend; compare TCP
2	Developing	0.19 to 0.48	15.0 to 18.0	23.7 to 26.0	0.6 to 1.5	0.58 to 0.77	normal, rare warn near S 0.4	Plan inspection; schedule lubrication and backlash check
3	Warning	0.48 to 0.82	18.0 to 21.6	26.0 to 30.0	1.5 to 2.5	0.77 to 1.0	warn	Inspect at next stop
4	Alarm	0.82 to 1.0	21.6 to 23.5	30.0 to 32.3	2.5 to 3.0	1.0 to 1.125	alarm	Intervene; reduce duty
5	Beyond model	above 1.0	above 23.5	above 32.3 or any sample at or above 34	above 3.0	above 1.125	alarm	Critical; specialist

13.5 Worked classification examples

Table 25. Worked examples: window features to classification

Window (5 cycles)	M (Nm)	Peak per cycle (Nm)	A6 (Nm)	Sigma (Nm)	Classification	Reason
W1	13.1	21.4 to 22.6	0.12	0.46	Healthy (L0)	All features in healthy band
W2	14.2	22.6 to 23.4	0.45	0.50	Watch (L1), stage 1	M above 13.5 and A6 above 0.4 although no sample above 26
W3	16.8	24.4 to 25.6	1.0	0.68	Stage 2, early degradation, no platform limit crossed	Mean shift 3.8 Nm; platform status normal for all samples
W4	18.3	25.0 to 27.2	1.6	0.80	Warning (L2)	About half the cycles contain a sample at or above 26
W5	21.0	28.0 to 30.2	2.4	0.96	Warning, close to alarm	Few samples at 30 Nm; M 21.0 below the 21.6 alarm-equivalent
W6	22.9	30.4 to 33.5	2.9	1.10	Alarm (L3), stage 4	Peaks above 30 every cycle
W7	24.0	33.0 to 35.2	3.1	1.2	Critical (L4), stage 5	Sample at or above 34 Nm and M above 23.5
W8	13.0	21.5 to 22.8, but one sample 31 Nm	0.15	0.46 (0.9 with outlier)	Transient or glitch, not wear	Single outlier, next cycle normal, M and A6 healthy

13.6 Direct answers to the standard diagnostic questions

Table 26. Standard questions and direct answers for Sensor 1

Question	Answer
Is this sensor reading normal?	Yes if all samples of the last 5 cycles lie in 3.1 to 23.1 Nm, M is 12.5 to 13.5 Nm and A6 is at most 0.2 Nm, during production cycle at steady speed. Otherwise evaluate Sections 6 and 7.
Is the machine approaching an abnormal condition?	Yes if M exceeds 13.5 Nm, A6 exceeds 0.4 Nm, noise sigma exceeds 0.55 Nm, or the per-cycle peak exceeds 23.5 Nm, even when the status is normal.
What value should trigger a warning?	26 Nm on the per-cycle peak [P]. Early advisory at M 15 Nm or A6 0.6 Nm [I].
What value should trigger an alarm?	30 Nm on the per-cycle peak [P]. Critical at 34 Nm on any sample [I].
How long can an abnormal value persist before escalation?	Warn: 2 consecutive cycles to open. Alarm: 2 consecutive cycles to open and intervention within the shift. Critical: immediate. Alarm lasting more than 10 minutes escalates to critical [I].
What legitimately changes the reading?	Operating state, speed override, payload, warm-up, supply or drive issues, temperature (Sections 10 and 11).
Is a transient spike different from a sustained condition?	Yes. A spike lasts 1 sample, returns within 0.5 to 1 s and leaves M, A6 and noise unchanged. A sustained condition repeats at the same phase for 2 or more cycles and moves M or A6.

14. Threshold Rationale, Sensitivity and Calibration

14.1 Detection characteristics of the project limits

The table shows what the project limits detect at each severity, from 3000 simulated cycles per row (seeded, 16 samples per cycle, noise included) [D]. It answers how late the platform status is compared with the true state of the gear train, and why feature thresholds are needed.

Table 27. Per-cycle peak statistics and limit detection by severity [D]

Severity r	Mean cycle peak (Nm)	5th to 95th percentile of cycle peak (Nm)	Cycles with a sample at or above 26 Nm	Cycles with a sample at or above 30 Nm	Mean cycle minimum (Nm)	Status seen by platform
0 (healthy)	21.7	20.9 to 22.4	0 percent	0 percent	4.3	normal
0.10	22.5	21.7 to 23.3	0 percent	0 percent	5.6	normal
0.25	23.8	22.8 to 24.8	0 percent	0 percent	7.4	normal
0.40	25.1	24.2 to 26.2	9 percent	0 percent	9.2	mostly normal, occasional warn
0.48	25.9	24.9 to 27.0	41 percent	0 percent	10.1	warn in about 4 of 10 cycles
0.50	26.1	25.1 to 27.2	54 percent	0 percent	10.4	warn flickering
0.60	27.1	26.2 to 28.2	97 percent	0 percent	11.4	warn each cycle
0.75	28.8	27.8 to 30.0	100 percent	5 percent	12.9	warn, rare alarm
0.82	29.7	28.6 to 30.9	100 percent	31 percent	13.5	warn, alarm in a third of cycles
0.90	30.7	29.5 to 32.0	100 percent	83 percent	14.2	alarm in most cycles
1.00	32.0	30.7 to 33.4	100 percent	100 percent	15.0	alarm each cycle

14.2 Why the thresholds sit where they do

Table 28. Threshold rationale

Threshold	Value	Rationale	Margin or sensitivity	Basis
warnHigh	26 Nm	Declared by the machine as an illustrative demo constant; sits about 4.25 Nm above the healthy noise-free peak (9.4 healthy sigma)	Zero healthy false warnings over 600 s (test); crossed at S 0.48	[P]
alarmHigh	30 Nm	Declared limit; 8.25 Nm above healthy peak	Crossed at S 0.82; about 79 percent of modeled severity range used	[P]
Critical 34 Nm	34 Nm	Above the noise-free full-severity peak (32.28) plus about 1.5 sigma of full-severity noise (1.125 each)	Roughly the upper 5 percent of full-severity cycle peaks	[I]
Healthy upper expected 23.1 Nm	23.1 Nm	Healthy per-cycle maximum observed 23.03 in 2000 cycles	Any sample above this is not produced by the healthy model	[D]
Mean band 12.5 to 13.5	+/- 0.5 Nm	About 10 sigma of the 5-cycle mean (0.052) so that false drift alerts are negligible, and wide enough to allow 0.5 Nm real-robot repeatability	Corresponds to S +/- 0.05	[I]
A6 watch 0.4 Nm	0.4 Nm	Above the 95th percentile of the one-cycle healthy estimate (0.39) and about 2x the five-cycle 95th percentile (0.18)	Equals S 0.13	[I]
A6 warning 1.5 Nm	1.5 Nm	Model value at S 0.5	-	[D]
Dwell 2 cycles (16 s)	16 s	Matches the planned 10 s dwell rounded up to whole cycles, since cyclic peaks cannot persist for 10 s	Avoids a single cycle outlier opening an alert	[I]
Clear margin -1 Nm	1 Nm	Hysteresis larger than the standard deviation of the cycle peak (0.4 to 0.7 Nm)	Prevents chattering at a limit	[I]

14.3 Calibrating the thresholds for a real robot

The values in this document are tuned to the simulated torque model. To transfer them to a real axis 4 the following adjustments are needed [E]; each is expressed as a rule so that a later revision can replace the numbers without changing the structure.

- Replace the healthy baseline: M, template peak, template minimum and sigma come from the baseline procedure of Section 13 on the real robot.
- Re-express limits as fractions of the axis rated torque if known: the project warn and alarm limits are 26 and 30 Nm, and the meaning of those values relative to the real rated torque must be established. As a general engineering practice, sustained torque above about 80 to 90 percent of the continuous rated torque of the motor and gearbox combination is treated as warning and above 100 percent as alarm [E].
- Scale the feature bands: keep the stage definitions in terms of severity S, and re-derive $M_S = M_{\text{baseline}} + S \times \Delta M_{\text{full}}$ where ΔM_{full} is the mean shift observed at end-of-life of a gearbox of the same type. For the simulator ΔM_{full} is 10.5 Nm.
- Re-estimate noise: real controller torque noise is usually a larger fraction than 3.5 percent of the mean; update the A6 and sigma thresholds from the measured baseline distribution (use the 95th percentile of baseline for the watch level).
- Verify the cycle period: the 8 s weld cycle is a simulator constant; on a real line the cycle time varies with part flow and the phase must be taken from the robot program counter or the weld gun signal.

15. Interpretation Guidelines

15.1 Decision rules for reading Sensor 1

RULE 1: IF all samples of the last 5 cycles lie between 3 and 23.1 Nm AND M lies between 12.5 and 13.5 Nm AND A6 is at most 0.2 Nm THEN the reading is NORMAL; no action.

RULE 2: IF M is between 13.5 and 15.0 Nm OR A6 is between 0.4 and 0.6 Nm for 5 consecutive cycles AND the robot is in production cycle at steady speed THEN classify as WATCH (early drift) and request the TCP deviation trend.

RULE 3: IF the per-cycle peak is at or above 26 Nm for 2 consecutive cycles THEN classify as WARNING, abnormal operation; evaluate the gearbox-wear signature in Document 02 and open an inspection request for the next planned stop.

RULE 4: IF the per-cycle peak is at or above 30 Nm for 2 consecutive cycles THEN classify as ALARM; recommend intervention and reduced duty; do not run to failure.

RULE 5: IF any sample is at or above 34 Nm, OR M is at or above 23.5 Nm, OR the alarm persists for more than 10 minutes THEN classify as CRITICAL and escalate to a maintenance specialist.

RULE 6: IF a high torque is observed AND the robot is not in production cycle at steady speed THEN do not apply thresholds; state that the operating context prevents classification.

15.2 Common misreadings

Table 29. Misreadings to avoid

Misreading	Why it is wrong	Correct reading
Mean 23.5 Nm is below 26, so the axis is healthy	Limits apply to peaks; at full severity the mean is 23.5 Nm while peaks reach 32.3 Nm [D]	Use M, A6 and noise as well as the limits
Status normal at this moment, so no fault	Status flickers each cycle at moderate severity; a normal sample can occur while the cycle is abnormal	Use cycle windows
Torque warning without TCP change means gearbox wear	In the model TCP rises first (r 0.27 for warn)	Check TCP trend; absence suggests other cause
A spike of 10 Nm in 1 s is a fault	Healthy steps reach 7 Nm per 0.5 s and slopes reach 12.4 Nm/s [D]	Compare with the cycle template for that phase
Low torque during the weld window means the sensor failed	Healthy torque there is 4 to 13 Nm [D]	Check phase before judging low values

16. Alert Escalation Logic

Table 30. Alert escalation table for Sensor 1

Level	Entry condition	Persistence to open	Clear condition	Owner and action	Time target [I]
L0 Normal	All features in healthy band	-	-	None	-
L1 Watch	M 13.5 to 15.0 Nm or A6 0.4 to 0.6 Nm or sigma above 0.55	5 cycles (40 s)	Features healthy for 5 cycles	Agent logs observation; compare with TCP	Review within shift
L2 Warning	Per-cycle peak at or above 26 Nm	2 cycles (16 s)	Peak below 25 Nm for 2 cycles	Open one investigation per machine (10 min cooldown)	Inspection at next planned stop
L3 Alarm	Per-cycle peak at or above 30 Nm	2 cycles (16 s)	Peak below 29 Nm for 2 cycles	Recommend intervention, reduced duty, stop planning	Intervention within the current shift [I]
L4 Critical	Sample at or above 34 Nm, M at or above 23.5 Nm, or alarm longer than 10 min	1 cycle confirmation	Only after inspection and verification run	Escalate to maintenance specialist; stop at safe point	Immediate
LS Sensor suspect	Flatline, saturation at 45, negative-looking values (0), or residual spikes	3 cycles	Data restored and verified	Investigate data path before mechanical work	Within shift

Table 31. Mapping to platform status [P] and to the planned agent

Platform behaviour today	Planned detector (blueprint)	This document
Per-sample warn at 26, alarm at 30, no hysteresis, steady amber or pulsing red on twin	Threshold with hysteresis (10 s open, 10 s clear), drift (z above 3 for 30 s, or slope over 10 minutes), debounce (10 min cooldown)	Apply the dwell to the per-cycle peak envelope and add feature-based drift levels
Marker triangle for warn, square for alarm	One open investigation per machine	L1 to L4 ladder and LS sensor-suspect level

17. Diagnostic Implications

Table 32. Diagnostic implications of Sensor 1 states

Sensor 1 state	Primary hypothesis	What to retrieve next	Corroboration needed
Healthy (L0)	No fault in axis 4 drive train	Nothing	None
Mean up, ripple at 0.75 Hz, noise up (L1 to L4)	GEARBOX_WEAR: damaged gear mesh in axis 4	Document 02 section on gearbox wear; Document 03 rules	TCP deviation rising, gun temperature unchanged
Mean up, no ripple, noise unchanged	Load change, lubrication or friction, payload change	Document 02 alternatives	Program, payload, temperature context
Step change in mean	Process change, collision, sensor offset	Document 02, Document 05	Event log, TCP step
Flat or saturated trace	Sensor or data path fault	Document 04 data-path table	seq gaps, stale status, diag fields
Intermittent peaks at one phase only	Local mechanical or program event	Document 05 intermittent case	Repeats at same phase?

The five documents are complementary. Document 01 gives normal values and levels; Document 02 explains how each fault changes them; Document 03 gives the procedure and decision matrices; Document 04 gives the maintenance actions; Document 05 gives worked cases. Only the Sensor 1 signature of GEARBOX_WEAR is a project-specific fault signature [P]; every other fault signature in the suite is engineering reference [E] and is labelled as such.

18. Summary, Safety Notes, References and Revision

18.1 Summary of key thresholds

Table 33. Key thresholds at a glance

Item	Value	Unit	Basis
Normal sample range	3.1 to 23.1	Nm	[D]
Healthy cycle mean	13.0 (12.5 to 13.5)	Nm	[P][I]
Healthy cycle peak	21.75	Nm	[D]
warnHigh (warning)	26	Nm	[P]
alarmHigh (alarm)	30	Nm	[P]
Critical (proposed)	34	Nm	[I]
Catalog range	0 to 45	Nm	[P]
Mean at which warn peak begins	18.0 (S 0.48)	Nm	[D]
Mean at which alarm peak begins	21.6 (S 0.82)	Nm	[D]
A6 watch / warning / alarm	0.4 / 1.5 / 2.5	Nm	[I]
Persistence to open warn or alarm	2 consecutive cycles (16 s)	-	[I]
Cycle period	8	s	[P]

18.2 Safety and escalation notes

This document is a prototype knowledge base and does not replace the robot manufacturer's manual, the plant's lock-out and tag-out procedure or site safety rules. Maintenance text must carry an OEM-procedure and lock-out caveat. The diagnostic agent cannot act on the machine: its outputs are reports and user-interface commands only. Escalate to a human specialist for any critical-level condition, any alarm that does not clear after intervention, and any case where the sensor itself is suspect.

18.3 References and engineering basis

- Repository: firmware/catalog.json (limits, range, scenarios), firmware/cdo-edge-gateway/simulation.h and server/scripts/lib/sim-models.mjs (equations), docs/mqtt-contract.md (topics, payloads, fault table), server/src/utls/channelStatus.js (status logic), server/test/simulator.test.js (asserted behaviour), docs/phase5-agent-blueprint.md (planned detector), docs/demo-binding-cheatsheet.md (mesh suggestion).

- Engineering practice: ISO 10816 / ISO 20816 style trend-based condition monitoring (compare against own baseline, trend before limits); ISO 9283 and ISO 10218 for industrial robot performance and safety context; general servo and gearbox diagnostic literature on current-signature analysis. These are cited as general background only; no clause-level compliance is claimed.
- Derived numbers were computed by replicating the repository equations in Python with a seeded random generator; percentages of samples in warn or alarm depend on the seed.

18.4 Revision notes

Table 34. Revision history

Rev	Date	Change
1.0	2026-10-08	Initial release for Sensor 1 of Robot Weld 01. Project values from CONTEXT_SENSOR1.md; extended levels labelled [E] and [I].